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A novel herbal preparation desensitizes mesenteric afferents to bradykinin in the rat small intestine

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Abstract *Herbal preparations are evolving as promising agents for the treatment of functional gastrointestinal disorders which are considered to be secondary to visceral hypersensitivity. We aimed to determine whether a new combination of six herbal extracts reduces the sensitivity of intestinal afferents in rat. Male Wistar rats (250–350 g, n = 6 per group) were gavaged with either vehicle or 2.5, 5 or 10 mL kg*$^{-1}$ *of STW 5-II, a herbal preparation which contains six extracts. Two hours later, animals were anaesthetized and extracellular multi-unit mesenteric afferent nerve recordings were obtained in the proximal jejunum in vivo. Afferent discharge to 5-hydroxytryptamine (5-HT) (5, 10, 20 and 40 μg kg*$^{-1}$*, i.v.), luminal distension (0–60 mmHg) and bradykinin (BK) (15, 30 and 60 μg kg*$^{-1}$*, i.v.) was recorded. At baseline, spontaneous afferent discharge was not different following pretreatment with the various doses of STW 5-II compared with vehicle. The pressure-dependent increase in afferent discharge to intraluminal ramp distension and the dose-dependent increase in afferent firing following 5-HT were also uninfluenced by STW 5-II pretreatment. In contrast, the afferent nerve responses to 15, 30 and 60 μg kg*$^{-1}$ *of BK were reduced following 10 mL kg*$^{-1}$ *STW 5-II with peaks at 106 ± 19, 153 ± 22 and 156 ± 25 imp s*$^{-1}$ *compared with 160 ± 15, 228 ± 14 and 220 ± 16 imp s*$^{-1}$ *following vehicle pretreatment (mean ± SEM, P < 0.05). Intestinal afferent sensitivity to BK which plays a prime role in nociception was reduced following STW 5-II. Thus, STW 5-II may be of therapeutic use for conditions that involve neuronal hypersensitivity and the release of BK in the intestine.*

Keywords *afferent, herbal medicine, jejunum, phytomedicine, visceral hypersensitivity.*

INTRODUCTION

The prevalence of functional bowel disorders is extremely high. It is estimated that about one third of the population is suffering from heartburn but only <15% of these patients have morphological alterations in the upper gastrointestinal tract.[1] The remaining patients have symptoms that are in large part secondary to functional disorders such as non-erosive reflux disease or functional dyspepsia. In the lower gastrointestinal tract, irritable bowel syndrome is the prevailing functional disorder which has similar epidemiological importance.[2] These disturbances in the upper and lower gastrointestinal tract frequently entail that quality of life is severely impaired.[3]

Visceral hypersensitivity is considered to be the prime mechanism that underlies these functional disorders in the gastrointestinal tract. The generally accepted concept to explain these conditions is that stimuli from the gastrointestinal tract are consciously perceived that would remain unrecognized in healthy individuals.[4–6] Based on these considerations, medical strategies were developed to reduce intestinal afferent sensitivity with the aim to improve patients' symptoms. However, the

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pharmacological approach to address well-defined targets such as the 5-HT_3 receptor turned out to be problematic as selective 5-HT_3 receptor antagonists had a rare but extremely severe side effect i. e. bowel ischaemia,[7] although it was questioned whether this potentially lethal condition is secondary to the drug or rather a potential consequence of the underlying disorder.[8]

It is not surprising, therefore, that the use of herbal preparations such as STW 5 (Iberogast®; Steigerwald GmbH, Darmstadt, Germany) which contains a combination of nine different herbal extracts, raised tremendous interest. The efficacy of this herbal preparation was shown in experimental and clinical studies, while unwanted side effects were virtually absent.[9–12] One intriguing observation was that STW 5 has the potential to reduce intestinal afferent sensitivity to chemical and mechanical stimuli in animal experiments.[13] This suggests that it may reduce visceral hypersensitivity and thereby specifically address the prime pathomechanism which appears to be responsible for the generation of symptoms in patients with functional bowel disorders.

In the present study, we aimed to characterize the action of a novel preparation containing six extracts (STW 5-II) on intestinal afferent sensitivity in the rat. We specifically hypothesized that this preparation reduces mesenteric afferent sensitivity to painful stimuli dose dependently.

METHODS

Herbal preparation

The herbal preparation STW 5-II contains six different extracts: bitter candytuft (Iberis amara totalis), peppermint leaves (Menthae piperitae folium), chamomile flowers (Matricariae flos), caraway fruits (Carvi fructus), melissa leaves (Melissae folium) and liquorice roots (Liquiritae radix). Components were chosen based on empirical knowledge and the herbal preparation was named STW 5-II. In this study it was administered at the same dose as STW 5 in previous studies [13] i.e. at 10 mL kg^{-1}, however, to provide insight in a potential dose–response relationship, it was also administered at 2.5 and 5 mL kg^{-1}. The solution was diluted when given at lower doses so that all animals received the same volume orally (10 mL kg^{-1}). The investigator was blinded for the dose of STW 5-II/vehicle that was administered by gavage.

Animals

Experiments were performed on male Wistar rats (300–400 g) that were held under standardized conditions (12 h/12 h dark/light cycle; regular rat chow; free access to tap water). Animals were fasted overnight before they received oral STW 5-II (2.5, 5 or 10 mL) or vehicle at random (vehicle: 30.8% ethanol/69.2% water). Each subgroup consisted of six animals. The animals were gavaged with STW 5-II or vehicle with a metal cannula which was introduced via the animals oesophagus in the stomach. The animal experiments were approved of by the appropriate institution prior to the study (Regierung Oberbayern).

Technique of afferent nerve recordings

Two hours after gavage, animals were deeply anesthetized with pentobarbitone (dose 60 mg kg^{-1} i.p.; Nembutal®, Sanofi Santé Animale, Libourne-Cedex, France). Via a neck incision, the trachea was cannulated in order to facilitate breathing. The left carotid artery was cannulated and connected to a pressure transducer in order to monitor arterial blood pressure. Furthermore, the left internal and external jugular veins were cannulated separately for the i.v. administration of mediators and to maintain anaesthesia by continuous pentobarbitone perfusion (0.5–1 mg kg^{-1} h^{-1}, i.v.). Animals were laparotomized and the abdominal wall sutured to a metal ring in order to create a well. A 5 cm loop of proximal jejunum was cannulated for intraluminal pressure recordings. Pressure was recorded from the proximal cannula through a separate canal that was connected to a pressure transducer (Ohmeda, DTX Plus Transducer, Ohmeda, NJ, USA), while the distal cannula remained open to the atmosphere except for ramp distensions of the intestinal loop. Absorption of fluid from the intestine was balanced by perfusion of normal saline into the loop at a rate of 1 mL h^{-1}. The intraluminal pressure was typically 2–3 mmHg at baseline. The abdomen was filled with light liquid paraffin and a neurovascular bundle from the mesentery of the proximal jejunum was placed and dissected on a black perspex platform. Multi-unit mesenteric afferent nerve recordings were obtained by cutting the dissected mesenteric nerve proximally and placing the distal end on an extracellular electrode, while a strip of connective tissue was placed on a second differential electrode as has been described.[14] The actual afferent recording started approximately 4 h following STW 5-II or vehicle gavage, as the preparation was started 2 h after administration and required typically 1.5 h. An additional 30 min period was required for signal stabilization.

The electrodes were connected to a CED single channel 1902 preamplifier/filter [Cambridge Electronic Design (CED), Cambridge, UK]. The signal was amplified 10 000 times and filtered with a bandwidth of 100 Hz to 1 kHz. Intestinal motor events and arterial blood pressure were monitored by recording the pressure via pressure transducers (see above) that were connected to another CED single channel 1902 preamplifier/filter. The output was saved on a hard drive and viewed online by Spike 2 software (version 4.01; CED) which was running on a personal computer.

Protocol for afferent nerve recordings

Once a stable recording had been established for 30 min, the intestinal afferent nerve discharge to 5, 10, 20 and 40 μg kg^{-1} 5 hydroxytryptamine (5-HT) i.v. was recorded. Mechanical ramp distension up to 60 mmHg was performed by filling the cannulated jejunal loop with neutral buffered saline (1 mL min^{-1}) through a perfusion pump (IVAC 711; IVAC Corp., San Diego, CA, USA). Finally, bradykinin (BK) was administered i.v. at 10, 20 and 40 μg kg^{-1}. Afferent discharge was recorded and the signal was saved online approximately 30 s before each stimulus was administered and continued for at least 1 min after blood pressure and afferent discharge frequency had come back to the prestimulus baseline level. Stimuli were administered at a minimum interval of 5 min. The relationship between afferent nerve discharge, arterial blood pressure and intestinal motility following 60 μg kg^{-1} of BK i.v. was studied in detail in a separate set of six experiments due to limitations in the quality of several arterial blood pressure and motility tracings in the STW 5-II experimental series.

Drugs

5 hydroxytryptamine and BK were purchased from Sigma Chemicals (St. Louis, MO, USA). Solutions were prepared and diluted in normal saline. Identical volumes (0.1 mL kg^{-1}) were injected i.v. independent of the dose administered. STW 5-II was provided by Steigerwald GmbH (Darmstadt, Germany) as a liquid preparation containing 30.8% ethanol.

Data analysis

Data were analysed with the investigator blinded for the administered dose of STW 5-II or vehicle. Baseline discharge frequency (imp s^{-1}) was calculated by averaging afferent nerve discharge per second for the 30 s period immediately prior to administration of the test stimuli. The response to a particular stimulus is expressed as the peak increase in the number of afferent impulses per second above baseline determined over a period of 3 s. In a preliminary analysis, differences were calculated by two-way ANOVA and *post hoc* Dunnetts test. This calculation was double-checked by a statistician (UM) with a more refined statistical analysis which identified the same differences in the data that are detailed in the results section: Linear mixed effects models were used to test the hypothesis that response profiles are identical under increasing stimuli for each dose group of STW 5-II, as this approach fitted our experimental setup and data more accurately when compared with standard ANOVA. Each animal was allowed to have an individual response level and individual response to the stimulus according to an overall group pattern. The zero hypothesis 'for each dose group of STW 5-II the response profile to the stimulus' was tested by a likelihood ratio test,[14] which is a basic tool of statistical interference and used to prove that a set of variables is needed to describe an observation appropriately and that simplifying the set of variables by removing components results in a systematic loss of information. This technique was used to determine whether the response profiles of the different stimuli are not equal for the different doses of STW 5-II. When significant, this is translated into the statement that the dose of STW 5-II alters the afferent nerve response to the appropriate stimulus. We used mixed (random) effect models to separate different sources of variation in the data set: measurement error and individual behaviour of the experimental unit during multiple measurements which is a common approach for an efficient analysis of small data sets. Data are presented as mean ± SEM. A probability of $P < 0.05$ was considered statistically significant.

RESULTS

Afferent discharge at baseline was 17.1 ± 1.6 imp s^{-1} in animals receiving STW 5-II at a dose of 10 mL kg^{-1} ($n = 6$) which was not different from 19.7 ± 2 imp s^{-1} in controls ($n = 6$). Afferent discharge at baseline was not different among the other subgroups receiving 2.5 mL and 5 mL kg^{-1} STW 5-II (data not shown).

5 hydroxytryptamine was administered at 5, 10, 20 and 40 μg kg^{-1} i.v. A dose-dependent increase in afferent nerve discharge was observed which was not different between vehicle controls and the groups receiving pretreatment with STW 5-II at the doses mentioned. At 40 μg kg^{-1} of 5-HT peak afferent discharge was 106 ± 17 imp s^{-1} following vehicle and 92 ± 24 imp s^{-1} following oral administration of 10 mL kg^{-1} STW 5-II (ns, Fig. 1A and B). The response profile to 5-HT and its components were not different in vehicle and STW 5-II

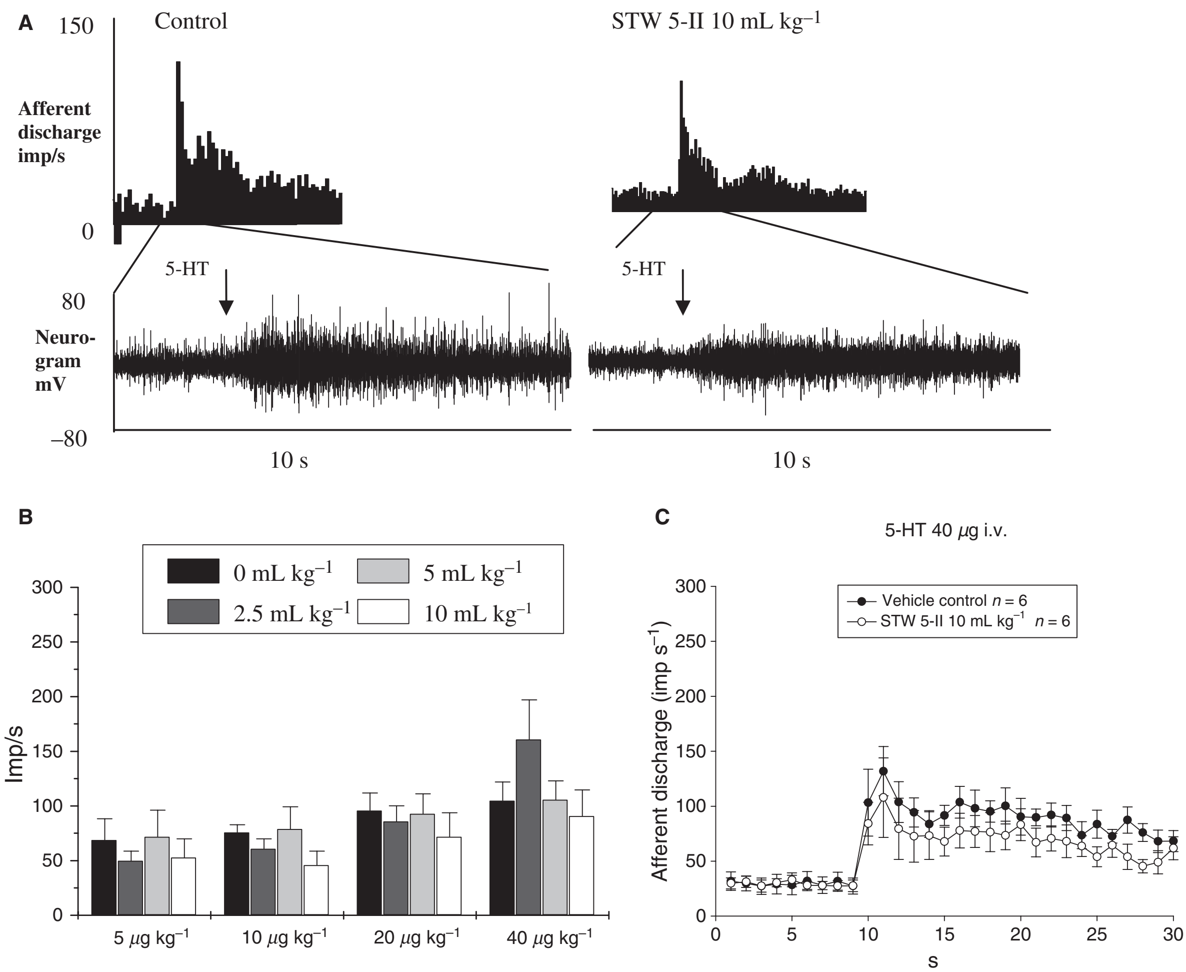

Figure 1 Panel (A): The upper trace shows sequential rate histograms of the mesenteric multi-unit afferent response to 5-HT at a dose of 40 μg kg^{-1} (i.v.) in a vehicle control animal and after STW 5-II (10 mL kg^{-1}; by gavage). The lower trace shows snap shots of a raw nerve signal at baseline and after stimulation with 5-HT. Panel (B): Group data are shown in this bar diagram (all n = 6). Peaks in mesenteric afferent nerve discharge following 5-HT i.v. are given as mean ± SEM. The different bar colours refer to the different doses of STW 5-II given as mL kg^{-1}. Panel (C): Mean afferent nerve responses to 5-HT (40 μg kg^{-1} i.v.) plotted as sequential rate histograms from six experiments each following vehicle or 10 mL kg^{-1} of STW 5-II. Standard error bars indicate SEM. Neither peak afferent nerve responses nor area under the curve analysis revealed a statistically significant difference (latter data not shown).

pretreated animals regardless of the dose administered. Figure 1(C) displays sequential rate histograms of the mean (SEM) afferent nerve discharge to 5-HT (40 μg kg^{-1}, i.v.) from vehicle and STW 5-II pretreated animals (10 mL kg^{-1}). An additional comparison of area under the curve did not reveal a statistically significant difference either (data not shown).

Mechanical ramp distension of the intestinal loop to 60 mmHg was followed by an increase in afferent firing in all subgroups. The afferent nerve response was not different in vehicle controls compared with the groups pretreated with the different doses of STW 5-II. Afferent nerve discharge peaked in controls at 234 ± 26 imp s^{-1} when the maximum intraluminal pressure of 60 mmHg was reached compared with 240 ± 42 imp s^{-1} after the highest dose of STW 5-II (n = 6, ns; Fig. 2A and B). Separate comparison of the low threshold component (0–20 mmHg) and the high threshold component (20–60 mmHg) of the afferent nerve response to ramp distension did not reveal a difference either when STW 5-II pretreated animals were compared to vehicle controls (Fig. 2C).

In control animals, afferent discharge was increased in a dose-dependent manner following administration of BK. The peak afferent nerve response was 160 ± 15 imp s^{-1} to 15 μg kg^{-1} of BK, 228 ± 14 imp s^{-1} to 30 μg kg^{-1} and 220 ± 16 imp s^{-1} to the 60 μg kg^{-1} dose. In animals pretreated with the highest dose of

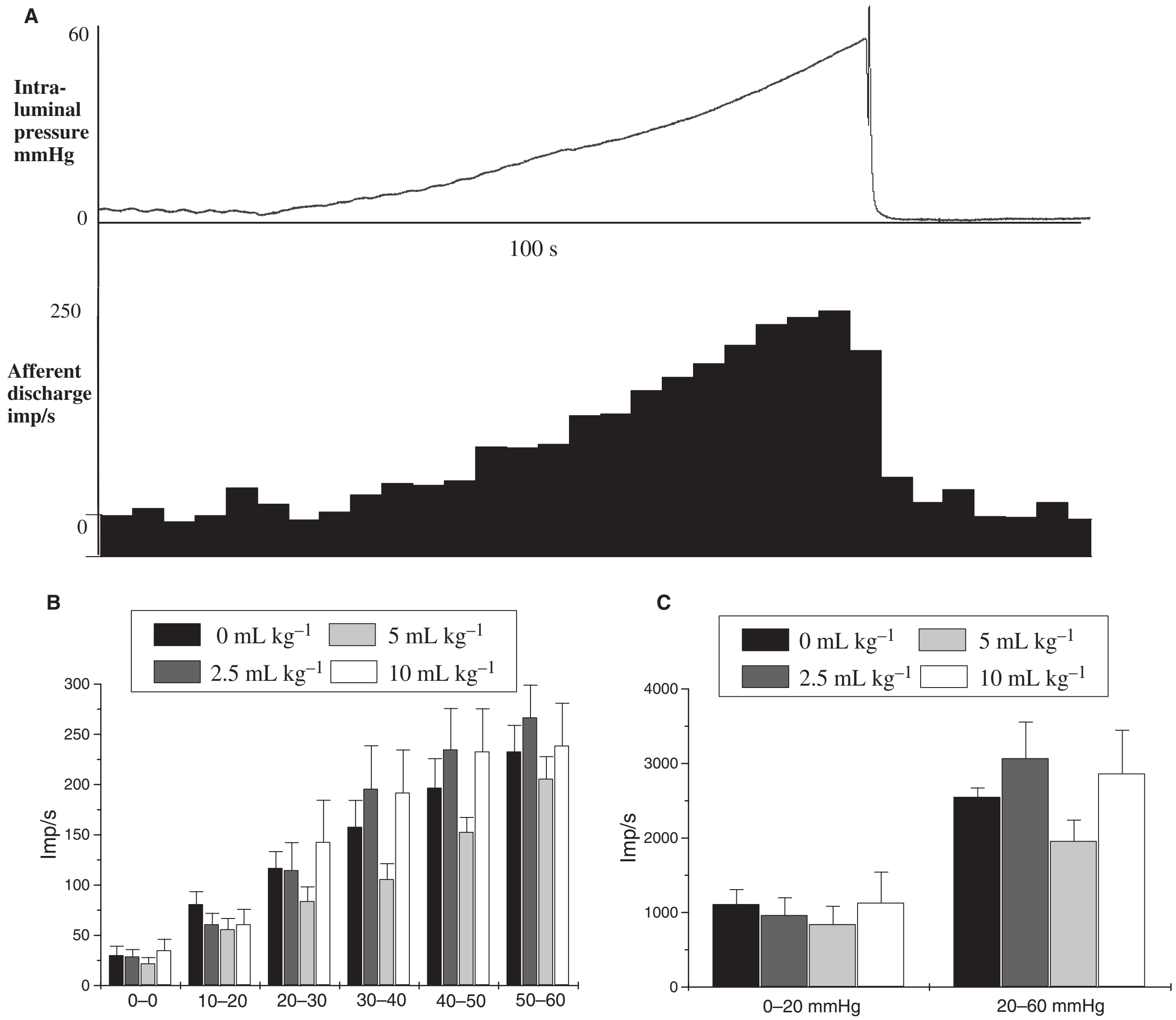

Figure 2 Panel (A): Representative afferent response to mechanical ramp distension to 60 mmHg in a vehicle control. The upper trace shows the intraluminal pressure curve. The lower trace displays the sequential rate histogram of a raw afferent nerve signal. Panel (B): Group data are shown in this bar diagram (n = 6 per group). Peaks in mesenteric afferent nerve discharge following ramp distension are given as mean ± SEM. Different bar colours refer to the doses of STW 5-II. Panel (C): The total number of afferent impulses determined as area under the curve minus baseline was additionally compared during low-threshold distension (0–20 mmHg) and high-threshold distension (20–60 mmHg) of the intestinal loop. Vehicle was compared to STW 5-II at 5 mL kg^{-1} as the lowest responses were seen with this dose. Differences in (B) and (C) were not statistically significant.

STW 5-II (10 mL kg^{-1}), peak afferent firing to 15, 30 and 60 μg kg^{-1} of BK was reduced to 106 ± 19, 153 ± 22 and 156 ± 25 imp s^{-1} ($P < 0.05$; Fig. 3A, C and D). The complete afferent nerve response to BK with all its components was different in vehicle and STW 5-II (10 mL kg^{-1}) pretreated animals which is illustrated in Fig. 3(B) by sequential rate histograms of the mean (SEM) afferent nerve discharge to BK (60 μg kg^{-1} i.v.) from vehicle and STW 5-II pretreated animals (10 mL kg^{-1}; each n = 6).

The relationship of the afferent nerve response to BK (60 μg kg^{-1} i.v.), arterial blood pressure and intestinal intraluminal pressure was studied in an additional set of experiments (n = 6). Administration of BK was accompanied by a fall in arterial blood pressure ($P < 0.001$) and a trend to a reduction of intestinal intraluminal pressure (ns). For details see representative recording in Fig. 4 and summarized data in Table 1.

DISCUSSION

In the present study, intestinal afferent nerve sensitivity to systemic BK was reduced following oral administration of the new herbal preparation STW 5-II at

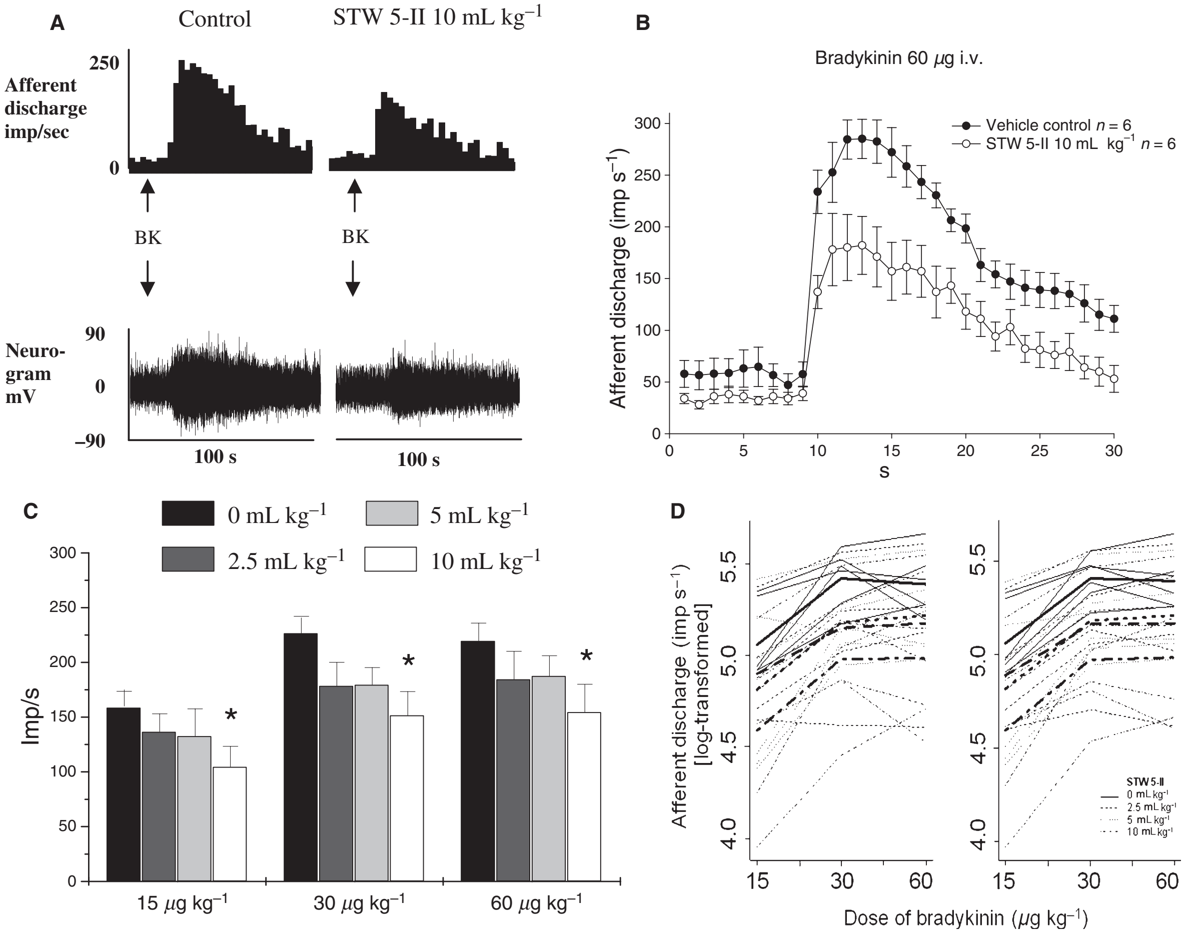

Figure 3 Panel (A): Representative recordings of the mesenteric afferent nerve response to bradykinin recorded from a vehicle control animal and after STW 5-II (10 mL kg^{-1} by gavage). The upper trace displays a sequential rate histogram after i.v. administration of bradykinin (60 μg kg^{-1}). The lower trace shows the corresponding raw nerve signal. Panel (B): Mean afferent nerve responses to bradykinin (60 μg kg^{-1}, i.v.) plotted as sequential rate histograms from six experiments each following vehicle or 10 mL kg^{-1} of STW 5-II. Standard error bars indicate SEM. Note that the complete response was reduced following pretreatment with STW 5-II. Panel (C): Group data are shown in this bar diagram. Peaks in mesenteric afferent nerve discharge following i.v. bradykinin are mean ± SEM (*$P < 0.05$, details of data analysis are given in the text). Panel (D): Data were analysed by random effect model and likelihood ratio test. (D) shows the profiles of the log-transformed response for each individual animal under increase of stimulus together with the mean log-transformed response profile for each dose group. The right panel shows the modelled data for each animal and the corresponding dose groups. Note that the model fits the data sufficiently good. The data shows significantly ($P = 0.0448$) that the response profile on the bradykinin stimulus is not equal for each dose of STW 5-II i.e. the effect of the 10 mL kg^{-1} dose is an overall decrease in response intensity, while the dose 2.5 mL kg^{-1} and the dose 5 mL kg^{-1} show no difference to the control group.

10 mL kg^{-1}, while mechanosensitivity and afferent sensitivity to 5-HT were uninfluenced. A trend to a reduced intestinal afferent nerve response was observed for 5-HT and ramp distension when analysed separately for low and high threshold distension of the intestinal loop, although this was beyond statistical significance regardless of whether the data was analysed by two-way ANOVA and *post hoc* Dunnetts test or the more refined approach with linear mixed effect models and likelihood ratio test. Thus, it cannot be completely ruled out that STW 5-II may have a minor effect on afferent sensitivity to these stimuli that was not delineated secondary to the limited number of experiments. As the herbal preparation contains ethanol (30.8%), vehicle with the same content was used as control. The vehicle is unlikely to have had any effect on afferent sensitivity, since the actual afferent recording started approximately 4 h following oral STW 5-II or vehicle administration. As adult rats quickly absorb and metabolize oral ethanol,[15] only

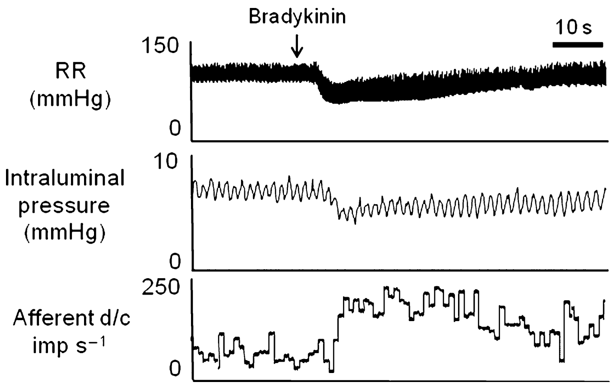

Figure 4 Representative recording of the intestinal afferent nerve response to bradykinin at 60 μg kg^{-1} (i.v.). The upper panel shows the systemic blood pressure response (arterial blood pressure recorded from a cannula in the left carotid artery). Intestinal intraluminal pressure is recorded in the middle panel. The lower panel displays sequential rate histograms of intestinal afferent nerve discharge per second at baseline and following bradykinin administration. Note the robust increase in afferent nerve discharge which is accompanied by a decrease in intraluminal and arterial blood pressure. The latter decrease was statistically significant (P < 0.001).

minor plasma concentrations of ethanol would persist in the animals during afferent nerve recordings.

Bradykinin is a 9-amino-acid peptide chain [16] generated from plasma during tissue damage and inflammation. Its effects are mediated via two G protein-coupled receptors, B_1 and B_2, with the latter being constitutive and the former induced by some cytokines and nerve growth factor.[17–19] The B_1-receptor is expressed as a result of tissue injury and was also described to play a role in inflammation.[17] It was further shown that activation of the kinin B_1-receptor recruits neutrophils via the chemokine CXCL5. This mechanism to attract neutrophils is used for example by endothelial cells.[18] It seems that during chronic inflammation, the role of the B_1-receptor in visceral nociception mechanisms becomes more dominant but probably not so much in naïve animals as investigated in our study.[7,16] In the latter animals, the B_2-receptor is more likely to mediate actions inflicted by BK.

Bradykinin sensitizes mainly spinal and not vagal afferents [20] with serosal terminals through an action on B_2-receptors and the BK-induced release of prostaglandins contributes to the overall magnitude of this response.[21–23] These findings corroborate animal studies showing that B_2-receptor antagonists attenuate visceral pain, at least in acute models in which inflammation is present.[19] The sensitization of spinal afferents by BK is in keeping with the concept that BK is a key player in the generation of pain perception [23–25] as spinal afferents fire predominantly when the gut is exposed to painful stimuli.[21,24,26]

When BK was administered systemically, a fall in arterial blood pressure was observed and a trend to decreased intraluminal pressure in the intestine. As there was no obvious change in intestinal motor events following BK, indirect activation of mechanosensitive afferents following intestinal pressure events elicited by BK is extremely unlikely. There was, however, a decrease of approximately 50% in arterial blood pressure following BK administration. In a previous study, we investigated the relationship between a drop in systemic blood pressure and intestinal afferent nerve discharge systematically.[27] We observed that a drop in blood pressure to a level below 50 mmHg (which was not reached in the present study) stimulates only a minor increase in afferent nerve discharge from ischaemically sensitive afferents. Furthermore, this afferent nerve response following a brisk decrease in arterial blood pressure was delayed by more than 50 s.[27] Thus, a minimal contribution of ischaemically sensitive afferents to the afferent nerve response to BK may have occurred but this does not explain the bulk of afferent firing to BK. Afferent discharge to systemic BK, therefore, is likely to stem from direct action of BK on afferent nerve terminals and does not appear to be secondary to systemic blood pressure drop or intestinal motor events. The afferent nerve response to BK was dose-dependent comparing the 15 and 30 μg kg^{-1} dose but a smaller increase was seen from the 30 to the 60 μg kg^{-1} dose. The latter doses were potentially on the high side of the dose–response curve which may

Table 1 Summarizes data on blood pressure, intestinal motor events and response latencies following 60 μg kg^{-1} of bradykinin i.v. in a separate experimental study in unpretreated animals (n = 6)

Systolic blood pressure (mmHg) at:	Baseline	123 ± 6.0
	Minimum following bradykinin i.v.	66 ± 4.3*
Intraluminal intestinal pressure (mmHg) at:	Baseline	7.2 ± 0.12
	Minimum following bradykinin i.v.	6.4 ± 0.50
Latencies (seconds) following bradykinin i.v. until:	Onset of afferent nerve response	7.25 ± 1.21
	Peak of afferent nerve response	14.4 ± 1.65
	Minimum in systolic blood pressure	8.1 ± 0.52
	Minimum in intraluminal pressure	10.5 ± 0.75

*P < 0.001 (comparison by paired t-test).

explain the moderate increase. This, however, had no consequences on the effect of STW 5-II at 10 mL kg^{-1} as it was present for all BK concentrations.

The mediator 5-HT was chosen for the protocol of the present study as it is extremely well characterized. Hillsley *et al.* demonstrated in the same experimental setup that 5-HT sensitizes vagal afferent nerve terminals in the intestinal mucosa via the 5-HT_3-receptor.[28,29] Although the i.v. administration of 5-HT is followed by a delayed intestinal motor response that also gives rise to some afferent firing, the bulk of the afferent nerve response occurs within few seconds after its administration and is clearly visible as an early peak of discharge frequency. In our study, quantification of peak afferent nerve discharge ensured that firing of directly stimulated 5-HT sensitive afferents was evaluated and compared. In contrast to the afferent nerve response to BK, modulation of the afferent nerve response to 5-HT following STW 5-II pretreatment was not found. One possibility is that BK and 5-HT act through different mechanisms. While BK binds to a G-protein coupled receptor,[30] 5-HT_3-receptors are ligand gated ion channels.[31] Possibly, ingredients of the STW 5-II preparation bind to G-protein coupled receptors and not to ligand gated ion channels which may explain the differential action. Other potential explanations are that STW 5-II alters binding of BK, function of the B_2 receptor or the chemical environment in the intestinal wall with the consequence that the action of BK was enhanced. Unfortunately, receptor binding studies of STW 5-II or its components are not available, so that these potential mechanisms cannot be further elucidated. Another possibility is that STW 5-II desensitizes certain subpopulations of mesenteric afferent nerves. Previous work clearly established that the BK response is mediated by spinal afferents and not by vagal or centrifugal fibres in the mesenteric nerve bundle.[20] If all spinal afferents were desensitized by STW 5-II, however, reduced afferent firing to high threshold distension would occur as discharge to this stimulus is also mediated by spinal afferents, while low threshold distension is predominantly mediated by vagal afferents.[32] As this was not observed, a generalized desensitization of spinal afferents by STW 5-II appears very unlikely.

In clinical studies, both STW 5 which contains nine herbal extracts and is commercially available and STW 5-II which is a research preparation with a combination of six extracts reduced symptoms in patients with irritable bowel syndrome and functional dyspepsia.[9,10,33,34] While both preparations were shown to provide a clinical benefit for the patients with functional dyspepsia and irritable bowel syndrome, their profile was different when investigated in our model of intestinal afferent nerve recording. In a previous study STW 5 reduced afferent sensitivity to all the stimuli tested which were 5-HT, mechanical ramp distension at low and high thresholds and BK.[13] As mentioned, no effect of STW 5-II was observed in the present study. It is of note, however, that comparison of these two studies is formally not correct. Nevertheless, it is intriguing that STW 5-II was shown to improve patients' symptoms, while the action was limited to BK in the present study.

Bradykinin is rarely mentioned in the context of the pathophysiology of functional bowel disorders[35,36] in contrast to 5-HT.[37] Given that STW 5-II works in patients but had an action in our experimental study that was limited to BK suggests that this inflammatory mediator may be more relevant for the understanding of functional bowel disorders than previously assumed. Alternatively, BK may simply be one mediator that is released during 'low-grade' inflammation which is a new hypothetical and unestablished concept which may help to explain the pathophysiology underlying irritable bowel syndrome.[38]

In summary, the agent STW 5-II has some potential for the treatment of functional gastrointestinal disorders. This is important as there is currently a lack of new and promising drugs to efficiently treat these conditions. STW 5-II seems to target specifically the transmission of chemical stimuli involved in pain generation from the gastrointestinal tract to the central nervous system (CNS) which are represented in a distorted fashion in patients suffering from functional bowel disorders.[39,40] In other words these stimuli seem to trigger a changed sensory gain of afferent neurons which are involved in the transmission process of visceral stimuli to the CNS[41] with the consequence that stimuli are exaggerated and out of the normal proportion of stimulus strength and subsequently become noxious.[36,42]

In conclusion, this study demonstrated that STW 5-II dose dependently reduces the intestinal afferent nerve response to BK which is a key mediator for the generation of nociception. This novel herbal preparation has the potential to reduce visceral hypersensitivity which is generally accepted as the prime pathomechanisms for functional bowel disorders.

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